



INTERNATIONAL ANTI-FOULING SYSTEM CERTIFICATE

(This certificate shall be supplemented by a Record of Anti-fouling Systems)

THE HONG KONG SPECIAL ADMINISTRATIVE REGION OF THE PEOPLE'S REPUBLIC OF CHINA

Issued under the International Convention on the Control of Harmful Anti-Fouling Systems on Ships under the authority of the Government of **THE HONG KONG SPECIAL ADMINISTRATIVE REGION OF THE PEOPLE'S REPUBLIC OF CHINA** by the Korean Register of Shipping.

When a Certificate has been previously issued, this Certificate replaces the certificate dated _____

Particulars of ship

Name of Ship	Distinctive Number or Letters	Port of Registry	Gross Tonnage	IMO Number
ALPINE MADELEINE	HK-2172 VREC6	HONG KONG	29,266	9380518

- (1) An anti-fouling system controlled under Annex 1 has not been applied during or after construction of this ship ☒
- (2) An anti-fouling system controlled under Annex 1 has been applied on this ship previously, but has been removed by _____ on _____ ☐
- (3) An anti-fouling system controlled under Annex 1 has been applied on this ship previously, but has been covered with a sealer coat applied by _____ on _____ ☐
- (4) An anti-fouling system controlled under Annex 1 was applied on this ship prior to _____ but must be removed or covered with a sealer coat prior to _____ ☐

1) In accordance with IMO ship identification number scheme adopted by the Organization by resolution.

2) Date of entry into force of then Control measure.

3) Date of expiration of any implementation period specified in article 4(2) or Annex 1.

Cert. No. SGP-IAFS-0053-16

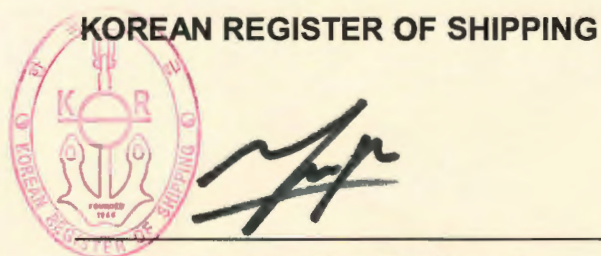
THIS IS TO CERTIFY THAT:

1. the ship has been surveyed in accordance with regulation 1 of Annex 4 to the Convention; and
2. the survey shows that the anti-fouling system on the ship complies with the applicable requirements of Annex 1 to the Convention.

Date of completion of the survey on which this certificate is issued : 10 July 2008

Issued at. Singapore

Date of issue 16 December 2016



Surveyor (Yap Eu Chen)

Cert. No. SGP-IAFS-0053-16

RECORD OF ANTI-FOULING SYSTEM

(This Record shall be permanently attached to the International Anti-Fouling System Certificate)

Particulars of ship

Name of ship : ALPINE MADELEINE

Distinctive number or letters : HK-2172 (VREC6)

IMO number : 9380518

Detail of anti-fouling system(s) applied

Type(s) of anti-fouling system(s) used :	Organotin-free self polishing type.....	☒
	Organotin-free ablative type.....	☐
	Organotin-free conventional type.....	☐
	Biocide-free silicon type paint.....	☐

Date(s) of application of anti-fouling system(s) : 16 May 2008

Name(s) of company(ies) and facility(ies)/location(s) where applied : Hyundai Mipo Dockyard Co., Ltd.

NO.3 Dry Dock of Hyundai Mipo Dockyard Co., Ltd.

Name(s) of anti-fouling system(s) manufacturer(s) : Chokwang Jotun Ltd.

Name(s) and colour(s) of anti-fouling system(s) : SEAFORCE 90, SEAQUANTUM CLASSIC

Light Red, Dark Red

Active ingredient(s) and their Chemical Abstract Services Registry Number(s) (CAS number(s)) :

Dicopper oxide, Bis(1-hydroxy-1h-pyridine-2-thionato-o,s) copper, Zinc Oxide

1317-39-1, 14915-37-8, 1314-13-2

Type(s) of sealer coat, if applicable :

Name(s) and colour(s) of sealer coat applied, if applicable :

Date of application of sealer coat :

THIS IS TO CERTIFY that this Record is correct in all respects.

Issued at Singapore

KOREAN REGISTER OF SHIPPING

Date of issue 16 December 2016



Surveyor (Yap Eu Chen)

Cert. No. SGP-IAFS-0053-16

Endorsement of the Records

THIS IS TO CERTIFY that a survey required in accordance with regulation 1(1)(b) of Annex 4 to the Convention found that the ship was in compliance with the Convention

Details of anti-fouling system(s) applied

Type(s) of anti-fouling system(s) used	Organotin-free self polishing type.....	☒
	Organotin-free ablative type.....	☐
	Organotin-free conventional type.....	☐
	Biocide-free silicon type paint.....	☐

Date(s) of application of anti-fouling system(s) 07 July 2013

Name(s) of company(ies) and facility(ies)/location(s) where applied DRYDOCK WORLD

DUBAI, UAE

Name(s) of anti-fouling system(s) manufacturer(s) HEMPEL A/S

Name(s) and colour(s) of anti-fouling system(s) HEMPEL'S ANTIFOULING GLOBIC 6000

BROWN & RED

Active ingredient(s) and their Chemical Abstract Services Registry Number(s) (CAS number(s))

copper(1) oxide(1317-39-1), copper pyrrhione(14915-37-8)

Type(s) of sealer coat, if applicable _____

Name(s) and colour(s) of sealer coat applied, if applicable _____

Date of application of sealer coat _____



Signed : *[Signature]* confirmed by Singapore Office

Place : DUBAI, UAE

Date : 07 July 2013

(Seal or stamp of the authority)

Endorsement of the Records

THIS IS TO CERTIFY that a survey required in accordance with regulation 1(1)(b) of Annex 4 to the Convention found that the ship was in compliance with the Convention

Details of anti-fouling system(s) applied

Type(s) of anti-fouling system(s) used	Organotin-free self polishing type.....	-
	Organotin-free ablative type.....	-
	Organotin-free conventional type.....	-
	Biocide-free silicon type paint.....	-

TBT-Free self-polishing type

Date(s) of application of anti-fouling system(s) 13 July 2018

Name(s) of company(ies) and facility(ies)/location(s) where applied Zhoushan Xinya Shipyard Co., Ltd
Zhoushan, China

Name(s) of anti-fouling system(s) manufacturer(s) Hempel Manufacturing(Malaysia) Sdn Bhd

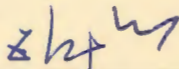
Name(s) and colour(s) of anti-fouling system(s) HEMPEL'S ANTIFOULING GLOBIC 9500M 78954
Red & Brown

Active ingredient(s) and their Chemical Abstract Services Registry Number(s) (CAS number(s))
copper(I) oxide, xylene, ethylbenzene, cupric oxide
1317-39-1, 1330-20-7, 100-41-4, 1317-38-0

Type(s) of sealer coat, if applicable _____

Name(s) and colour(s) of sealer coat applied, if applicable _____

Date of application of sealer coat _____

Signed : 

Place : Zhoushan, China

Date : 15 July 2018

(Seal or stamp of the authority)

